Lattice Gauge Theory on the Interaction Complex of a Viscous Particle System

Abstract

We construct a finite, explicit $SU(3)$ lattice gauge theory from the discrete trajectory data of a viscous multi-particle system. Starting from a recorded run of the interacticle algorithm, we extract a directed 2-complex whose nodes are spacetime particle states, whose edges encode causal propagation and pairwise viscous interactions, and whose triangular faces close the interaction-ahead boundaries. At each node we form a complex color triplet by modulating the viscous force vector with a kinematic phase factor of de Broglie type; these triplets encode momentum direction in the internal color space. Edge-to-edge parallel transport in $SU(3)$ is then defined by transporting color frames induced by the color triplets across oriented edges. Wilson loop traces, the standard plaquette action, and covariant fermionic and scalar matter actions are constructed on the same complex. The complete data package $\mathcal{Q}$ is proved to be finite and $SU(3)$ gauge-invariant. Continuum limit, path-integral quantization, and axiom verification are deferred to companion papers.

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1 Introduction

Lattice gauge theory, introduced by Wilson [5], provides a non-perturbative regularization of quantum field theories by replacing continuous spacetime with a discrete lattice and assigning $SU(N)$ group elements to the links. In the standard formulation the link variables are *independent integration variables* of a path integral, and the gauge group is imposed by hand. The present paper takes a different starting point: we ask whether a natural $SU(3)$ gauge structure can be *extracted* from the discrete trajectory data of an interacting classical system, without imposing any gauge group a priori.

The data source is the interacticle algorithm, a multi-particle sampler in which walkers move through a latent space under underdamped Langevin dynamics with a row-normalized viscous alignment force [1]. At each discrete timestep, each alive walker i experiences a force

$$\mathbf{F}_i^{\text{visc}} = \nu_{\text{visc}} \sum_{j \neq i} \omega_{ij}(v_j - v_i)$$

that pulls its velocity toward a weighted average of its interaction partners. This force is a vector in $\mathbb{R}^3$ and therefore lives naturally in the fundamental representation of $O(3)$.

The key observation is that $\mathbb{R}^3$ is three-dimensional, matching the defining dimension of the fundamental representation of $SU(3)$, the gauge group of the strong nuclear force [2]. By complexifying the viscous force with a kinematic phase factor analogous to the de Broglie phase $e^{ip\ell/\hbar}$, each spacetime node acquires a complex color triplet $\mathbf{c}_x \in \mathbb{C}^3$. The color triplets at two neighboring nodes define interaction data from which we build an $SU(3)$ parallel transport $U_3(e)$ for every edge e by a local frame construction. No additional input is required: the gauge group, the link variables, and the coupling strength all emerge from the dynamics of the viscous particle system.

In detail, we first build a deterministic $SU(3)$ frame at each node from the color triplet, then define U_3 as the frame transition between source and target nodes. The same data can be decomposed in the Gell-Mann basis after taking the matrix logarithm. The coupling parameter $g_3 > 0$ is placed to match the continuum normalization in $D_\mu = \partial_\mu - ig_3 A_\mu^a \lambda_a/2$.

This paper supplies all the formal definitions and proofs required for a subsequent axiom-verification paper. The main result is Theorem 8.1: the tuple $\mathcal{Q} = (\text{complex}, \{U_3(e)\}, S_{\text{W}}, S_{\text{F}}, S_\phi)$ is finite, explicit, and $SU(3)$ gauge-invariant.

Organization. Section 2 fixes notation, imports kernel parameters from the companion convergence paper, and tabulates all scalar parameters used throughout. Section 3 constructs the directed 2-complex $\mathcal{F} = (\mathcal{N}, \mathcal{E}, \mathcal{T})$ from the recorded run. Section 4 defines the viscous force on nodes and edges and proves its covariance under orthogonal transformations. Section 5 introduces color triplets, the Gell-Mann basis, and the $SU(3)$ edge transport. Section 6 defines path holonomies, plaquette holonomies, and the Wilson action and proves gauge invariance. Section 7 constructs covariant fermionic and scalar matter actions. Section 8 packages the complete data object and states the finiteness theorem. Sections 9 and 10 discuss connections to standard lattice QCD, the continuum limit, and quantization.

2 Notation and setup

Definition 2.1 (Kernel parameters and trajectory data)**.** *Let $d = 3$. The spatial dimension is fixed to three throughout because the viscous force is a vector in physical space and its three components form a natural triplet for the fundamental representation of $SU(3)$. Fix $N \geq 1$ and the index set*

$\mathcal{I} := \{1, \dots, N\}$. At each time every trajectory entry is recorded for all $i \in \mathcal{I}$. The alive set $\mathcal{A}(t) \subseteq \mathcal{I}$ records which indices are active. Fix kernel parameters

$$\ell_{\text{visc}} > 0, \qquad \kappa_0 > 0, \qquad \nu_{\text{visc}} \geq 0,$$

and define the regularized Gaussian kernel

$$K_\rho(x, y) := \kappa_0 + \exp\left(-\frac{\|x - y\|^2}{2\ell_{\text{visc}}^2}\right), \qquad \rho := \ell_{\text{visc}}.$$

The kernel satisfies

$$\kappa_* := \inf_{x,y} K_\rho(x, y) = \kappa_0 > 0, \qquad \kappa^* := \sup_{x,y} K_\rho(x, y) = 1 + \kappa_0 < \infty.$$

These bounds are imported from the companion paper on kinetic convergence; no other results from that paper are used here.

A recorded trajectory is available at discrete times

$$\mathbb{T} = \{0, 1, \dots, T\}, \quad T \in \mathbb{N},$$

with finite alive sets $\mathcal{A}(t) \subseteq \mathcal{I}$ and position, velocity, and fitness data

$$x_i(t),\, v_i(t) \in \mathbb{R}^3, \qquad \Phi_i(t) \in \mathbb{R}, \qquad i \in \mathcal{I},\ t \in \mathbb{T}.$$

At each timestep t, the finite set $\mathcal{P}_t \subseteq \mathcal{A}(t) \times \mathcal{A}(t) \setminus \{(i, i) : i \in \mathcal{A}(t)\}$ records the ordered interaction companions used by the algorithm.

The following table summarizes all scalar parameters introduced in this paper, together with their role and the constraint they must satisfy.

Symbol	Role	Constraint
ℓ_{visc}	Viscous kernel length scale	> 0
κ_0	Kernel floor (regularization)	> 0
ν_{visc}	Viscous coupling strength	≥ 0
g_3	Gauge coupling constant	> 0
ℓ_0	Length unit for de Broglie phase	> 0
$\hbar_{\text{eff}}$	Effective action quantum	> 0
μ_p	Momentum coupling to phase	> 0
ϵ_c	Color regularization	> 0
η	Distance regularization in scalar action	> 0
m_F	Fermion mass	> 0
m_ϕ	Scalar mass	> 0

Lemma 2.2 (Kernel orthogonal symmetry)**.** *For every orthogonal matrix $O \in O(3)$ and all $x, y \in \mathbb{R}^3$,*

$$K_\rho(Ox, Oy) = K_\rho(x, y).$$

Proof. By definition,

$$\|Ox - Oy\|^2 = \|O(x - y)\|^2 = \|x - y\|^2,$$

so the Gaussian factor is invariant under simultaneous orthogonal transformation. □

Lemma 2.3 (Orthogonal covariance of viscous forces)**.** *Let $O \in O(3)$ and define transformed data $x_i'(t) = Ox_i(t)$, $v_i'(t) = Ov_i(t)$. Then for each node $x = n_{i,t}$,*

$$\mathbf{F}^{\text{visc}}(x; \textit{transformed data}) = O\,\mathbf{F}^{\text{visc}}(x; \textit{original data}).$$

Proof. Lemma 2.2 gives $K_\rho(Ox_i, Ox_j) = K_\rho(x_i, x_j)$, hence

$$\deg'_{\text{IG}}(i,t) = \deg_{\text{IG}}(i,t), \qquad \omega_{ij}^{\text{IG}\prime}(t) = \omega_{ij}^{\text{IG}}(t).$$

For each edge $e = (n_{i,t} \to n_{j,t})$,

$$\mathbf{F}^{\text{visc}\prime}(e) = \nu_{\text{visc}}\,\omega_{ij}^{\text{IG}}(t)(Ov_j - Ov_i) = O\,\mathbf{F}^{\text{visc}}(e).$$

Summing outgoing IG edges at a fixed source node preserves left multiplication by O. □

3 Recorded run and interaction complex

Definition 3.1 (Recorded interacticle run)**.** *The recorded run is the tuple*

$$\mathscr{R} = \Big(N, \mathbb{T}, \{\mathcal{A}(t)\}_{t\in\mathbb{T}}, \{\mathcal{P}_t\}_{t\in\mathbb{T}}, \{x_i(t), v_i(t), \Phi_i(t)\}_{i\in\mathcal{I}, t\in\mathbb{T}}, \{K_\rho\}\Big).$$

Definition 3.2 (Node set and projections)**.** *For each pair $(i,t) \in \mathcal{I} \times \mathbb{T}$ define a spacetime node $n_{i,t}$ and set*

$$\mathcal{N} := \{n_{i,t} \mid i \in \mathcal{I},\ t \in \mathbb{T}\}.$$

Define spatial and velocity projections

$$\mathbf{x}(n_{i,t}) = x_i(t), \qquad \mathbf{v}(n_{i,t}) = v_i(t).$$

Both maps are well-defined by definition of $\mathcal{N}$.

Definition 3.3 (Oriented edge sets)**.** *Define three families of oriented edges:*

$$\begin{aligned} E_{\text{CST}} &:= \{(n_{i,t} \to n_{i,t+1}) : i \in \mathcal{A}(t),\ t = 0, \dots, T-1\}, \\ E_{\text{IG}} &:= \{(n_{i,t} \to n_{j,t}) : (i,j) \in \mathcal{P}_t,\ t \in \mathbb{T}\}, \\ E_{\text{IA}} &:= \{(n_{i,t+1} \to n_{j,t}) : (i,j) \in \mathcal{P}_t,\ t = 0, \dots, T-1\}. \end{aligned}$$

For convenience, write

$$e_{\text{CST}}(n_{i,t} \to n_{i,t+1}) := (n_{i,t} \to n_{i,t+1}), \quad e_{\text{IG}}(n_{i,t} \to n_{j,t}) := (n_{i,t} \to n_{j,t}), \quad e_{\text{IA}}(n_{i,t+1} \to n_{j,t}) := (n_{i,t+1} \to n_{j,t}).$$

By construction all source and target nodes of E_{IA} lie in $\mathcal{N}$ because $\mathcal{N}$ is defined for all $(i,t) \in \mathcal{I} \times \mathbb{T}$. The causal set tree (CST) edges E_{CST} encode temporal propagation: walker i at time t evolves forward to time $t+1$. The interaction graph (IG) edges E_{IG} encode same-time viscous couplings: walker i interacts with companion j at the same timestep. The interaction-ahead (IA) edges E_{IA} encode forward-in-time interactions: the updated state of i at $t+1$ is causally linked back to the companion j at t. Set

$$\mathcal{E} := E_{\text{CST}} \cup E_{\text{IG}} \cup E_{\text{IA}}.$$

For $e = (u \to v) \in \mathcal{E}$, write $s(e) := u$ and $t(e) := v$, and define $e^{-1} := (t(e) \to s(e))$.

Definition 3.4 (Interaction triangles and plaquettes)**.** *For each $(i,j) \in \mathcal{P}_t$ with $t < T$, define the oriented interaction triangle*

$$\triangle_{ij,t} := (n_{i,t}, n_{i,t+1}, n_{j,t}),$$

with oriented boundary

$$\partial \triangle_{ij,t} = e_{\mathrm{CST}}(n_{i,t} \to n_{i,t+1}) + e_{\mathrm{IA}}(n_{i,t+1} \to n_{j,t}) - e_{\mathrm{IG}}(n_{i,t} \to n_{j,t}).$$

If both ordered pairs $(i,j), (j,i) \in \mathcal{P}_t$ with $t < T$, define the plaquette

$$P_{ij,t} := \triangle_{ij,t} \cup \triangle_{ji,t}$$

with boundary 4-cycle

$$\partial P_{ij,t} = e_{\mathrm{CST}}(n_{i,t} \to n_{i,t+1}) + e_{\mathrm{IA}}(n_{i,t+1} \to n_{j,t}) + e_{\mathrm{CST}}(n_{j,t} \to n_{j,t+1}) + e_{\mathrm{IA}}(n_{j,t+1} \to n_{i,t}).$$

The IG contributions from the two triangles cancel in the plaquette boundary because they appear with opposite signs. Set

$$\mathcal{T} := \{\triangle_{ij,t} \mid (i,j) \in \mathcal{P}_t,\ t = 0, \dots, T-1\}, \qquad \mathcal{F} := (\mathcal{N}, \mathcal{E}, \mathcal{T}).$$

Define the underlying undirected structure

$$\bar{E}_{\mathrm{CST}} := \{\{u,v\} : (u \to v) \in E_{\mathrm{CST}}\}, \quad \bar{E}_{\mathrm{IG}} := \{\{u,v\} : (u \to v) \in E_{\mathrm{IG}}\}, \quad \bar{E}_{\mathrm{IA}} := \{\{u,v\} : (u \to v) \in E_{\mathrm{IA}}\},$$
$$\bar{E} := \bar{E}_{\mathrm{CST}} \cup \bar{E}_{\mathrm{IG}} \cup \bar{E}_{\mathrm{IA}}, \qquad \mathcal{F}_{\mathrm{und}} := (\mathcal{N}, \bar{E}, \mathcal{T}).$$

Proposition 3.5 (Directed 2-complex structure)**.** *$\mathcal{F}_{\mathrm{und}}$ is a finite undirected 2-complex. For each oriented edge $e = (u \to v) \in \mathcal{E}$ define $\partial_1 e = v - u$ (in the free $\mathbb{Z}$-module on $\mathcal{N}$), and for each triangle define*

$$\partial_2 \triangle_{ij,t} = e_{\mathrm{CST}}(n_{i,t} \to n_{i,t+1}) + e_{\mathrm{IA}}(n_{i,t+1} \to n_{j,t}) - e_{\mathrm{IG}}(n_{i,t} \to n_{j,t}).$$

For each $t < T$ with $(i,j) \in \mathcal{P}_t$,

$$\partial_1 \partial_2 \triangle_{ij,t} = 0,$$

and the plaquette boundary satisfies $\partial P_{ij,t} = \partial \triangle_{ij,t} + \partial \triangle_{ji,t}$, with the shared IG edges canceling by opposite orientations.

Proof. Each node, edge, and face is built from the finite sets of Definitions 3.1–3.4, so finiteness is clear. The boundary-of-boundary identity $\partial_1 \partial_2 = 0$ follows by direct computation:

$$\partial_1 \partial_2 \triangle_{ij,t} = (n_{i,t+1} - n_{i,t}) + (n_{j,t} - n_{i,t+1}) - (n_{j,t} - n_{i,t}) = 0.$$

In the sum defining $\partial P_{ij,t}$, the IG edge $e_{\mathrm{IG}}(n_{i,t} \to n_{j,t})$ from $\partial_2 \triangle_{ij,t}$ appears with coefficient -1, while $e_{\mathrm{IG}}(n_{j,t} \to n_{i,t})$ from $\partial_2 \triangle_{ji,t}$ is the reverse edge, hence it appears with coefficient $+1$ but represents the same undirected edge. In the oriented sum these contributions cancel, leaving only CST and IA edges. □

Proposition 3.6 (Finiteness and cardinality)**.** *All sets in Definitions 3.1–3.4 are finite. With $m_t := |\mathcal{P}_t|$ and*

$$E_{\mathrm{IG}}^{<T} := \{(n_{i,t} \to n_{j,t}) : (i,j) \in \mathcal{P}_t,\ t \in \{0, \dots, T-1\}\},$$

the cardinalities satisfy

$$|E_{\mathrm{CST}}| = \sum_{t=0}^{T-1} |\mathcal{A}(t)|, \qquad |E_{\mathrm{IG}}^{<T}| = |\mathcal{T}| = |E_{\mathrm{IA}}| = \sum_{t=0}^{T-1} m_t, \qquad |E_{\mathrm{IG}}| = \sum_{t=0}^{T} m_t.$$

Proof. Finiteness follows from finiteness of $\mathbb{T}$, all $\mathcal{A}(t)$, and all $\mathcal{P}_t$. The cardinality identities hold by one-to-one correspondence: each $(i,t) \in \bigcup_{t=0}^{T-1} \mathcal{A}(t) \times \{t\}$ contributes one CST edge, and each $(i,j,t) \in \bigcup_t \mathcal{P}_t \times \{t < T\}$ contributes one IG (pre-T), IA, and triangle. The last timestep $t = T$ contributes m_T additional IG edges. □

4 Viscous force from interaction edges

Definition 4.1 (IG degree and pairwise force coefficients)**.** *For each (i,t) define the raw IG degree and normalized weights*

$$\deg_{\mathrm{IG}}(i,t) := \sum_{(i,j)\in\mathcal{P}_t} K_\rho(x_i(t), x_j(t)),$$

$$\omega_{ij}^{\mathrm{IG}}(t) := \begin{cases} \dfrac{K_\rho(x_i(t), x_j(t))}{\deg_{\mathrm{IG}}(i,t)}, & (i,j)\in\mathcal{P}_t,\ \deg_{\mathrm{IG}}(i,t) > 0, \\ 0, & \textit{otherwise.} \end{cases}$$

Definition 4.2 (Viscous pair force and node force)**.** *For each edge $e = (n_{i,t} \to n_{j,t}) \in E_{\mathrm{IG}}$ define*

$$\mathbf{F}^{\mathrm{visc}}(e) := \nu_{\mathrm{visc}}\omega_{ij}^{\mathrm{IG}}(t)\,(v_j(t) - v_i(t)) \in \mathbb{R}^3.$$

For $x = n_{i,t} \in \mathcal{N}$ define

$$\mathbf{F}^{\mathrm{visc}}(x) := \sum_{\substack{e\in E_{\mathrm{IG}} \\ s(e)=x}} \mathbf{F}^{\mathrm{visc}}(e).$$

If the sum is over an empty set, it is 0.

Lemma 4.3 (Weight properties)**.** *For every $i \in \mathcal{A}(t)$ and $t \in \mathbb{T}$,*

$$\omega_{ij}^{\mathrm{IG}}(t) \geq 0, \qquad \sum_{j:(i,j)\in\mathcal{P}_t} \omega_{ij}^{\mathrm{IG}}(t) = \mathbf{1}_{\{\deg_{\mathrm{IG}}(i,t)>0\}}.$$

Proof. Positivity is immediate from $K_\rho \geq \kappa_* > 0$. If $\deg_{\mathrm{IG}}(i,t) > 0$, summing the definitions over j gives

$$\sum_{j:(i,j)\in\mathcal{P}_t} \omega_{ij}^{\mathrm{IG}}(t) = \frac{1}{\deg_{\mathrm{IG}}(i,t)} \sum_{j:(i,j)\in\mathcal{P}_t} K_\rho(x_i(t), x_j(t)) = 1.$$

If $\deg_{\mathrm{IG}}(i,t) = 0$, every summand is 0 by convention. □

Lemma 4.4 (Node reconstruction of viscous force)**.** *For every $x = n_{i,t} \in \mathcal{N}$,*

$$\mathbf{F}^{\mathrm{visc}}(x) = \sum_{\substack{e\in E_{\mathrm{IG}} \\ s(e)=x}} \mathbf{F}^{\mathrm{visc}}(e).$$

Proof. By Definition 3.3, the edges with source $n_{i,t}$ in E_{IG} are exactly $\{(n_{i,t} \to n_{j,t}) : (i,j) \in \mathcal{P}_t\}$, which is the same index set used in Definition 4.2. □

Proposition 4.5 (Orthogonal basis covariance)**.** *Let $O \in O(3)$ and consider transformed data $x_i'(t) = Ox_i(t)$, $v_i'(t) = Ov_i(t)$. For every node $x \in \mathcal{N}$,*

$$\mathbf{F}^{\mathrm{visc}}(x; \textit{transformed data}) = O\,\mathbf{F}^{\mathrm{visc}}(x; \textit{original data}).$$

Moreover, real orthogonal matrices satisfy $O^\top = O^{-1}$, hence $O^\dagger = O^{-1}$, so $O \in U(3)$. Consequently, orthogonal changes of the physical coordinate frame embed in unitary changes of the complexified color index.

Proof. Covariance of the force is Lemma 2.3. For the embedding: since O has real entries, $O^\dagger = O^\top = O^{-1}$, so $O^\dagger O = I$ and O is unitary when viewed as a complex matrix. □

5 Color triplets and $SU(3)$ edge transport

Definition 5.1 (Normalized color vectors)**.** *Fix constants $g_3 > 0,\ \ell_0 > 0,\ \hbar_{\text{eff}} > 0,\ \mu_p > 0,\ \epsilon_c > 0$. For $x = n_{i,t} \in \mathcal{N}$ and each coordinate $\alpha \in \{1, 2, 3\}$, define*

$$\tilde{c}_x^{(\alpha)} := (\mathbf{F}^{\text{visc}}(x))_\alpha \exp\left(i \frac{\mu_p\, v_i^{(\alpha)}(t) \ell_0}{\hbar_{\text{eff}}} \right),$$

$$\mathbf{c}_x := \frac{\tilde{\mathbf{c}}_x}{\sqrt{\tilde{\mathbf{c}}_x^\dagger \tilde{\mathbf{c}}_x + \epsilon_c^2}} \in \mathbb{C}^3.$$

Theorem 5.2 (Emergent local $SU(3)$ color symmetry)**.** *For any map $\Omega : \mathcal{N} \to O(3)$ and each node $x \in \mathcal{N}$, define*

$$\mathbf{c}_x^\Omega := \Omega(x)\mathbf{c}_x.$$

Because $\Omega(x)$ is real orthogonal, $\|\mathbf{c}_x^\Omega\| = \|\mathbf{c}_x\|$ for every $x \in \mathcal{N}$. Hence the color data carry a local $O(3)$ redundancy, which embeds as real unitaries into $U(3)$. Imposing $\det \Omega(x) = 1$ for every node yields a local $SO(3)$ subgroup $SO(3) \subset SU(3)$ on each node. The full local $SU(3)$ bundle symmetry used below is then defined explicitly by $\Omega : \mathcal{N} \to SU(3)$ in Definition 6.4.

Proof.

$$\mathbf{c}_x^\Omega = \frac{\Omega(x)\tilde{\mathbf{c}}_x}{\sqrt{\tilde{\mathbf{c}}_x^\dagger \tilde{\mathbf{c}}_x + \epsilon_c^2}} = \Omega(x)\mathbf{c}_x.$$

Since $\Omega(x)$ is orthogonal,

$$\|\mathbf{c}_x^\Omega\|^2 = \frac{\tilde{\mathbf{c}}_x^\dagger \Omega(x)^\dagger \Omega(x) \tilde{\mathbf{c}}_x}{\tilde{\mathbf{c}}_x^\dagger \tilde{\mathbf{c}}_x + \epsilon_c^2} = \frac{\tilde{\mathbf{c}}_x^\dagger \tilde{\mathbf{c}}_x + \epsilon_c^2}{\tilde{\mathbf{c}}_x^\dagger \tilde{\mathbf{c}}_x + \epsilon_c^2} = \|\mathbf{c}_x\|^2.$$

Therefore the normalization denominator is unchanged and $\|\mathbf{c}_x^\Omega\| = \|\mathbf{c}_x\|$. □

Remark 5.3 (Physical interpretation of the color construction)*.* The factor $\exp\big(i\mu_p v_i^{(\alpha)} \ell_0/\hbar_{\text{eff}}\big)$ is a discrete analogue of the de Broglie phase $e^{ip\ell/\hbar}$: it encodes the kinematic momentum of walker i in the α-direction into the phase of the α-th color component. Nodes moving in different directions therefore carry genuinely different internal phases, even if they experience the same force magnitude. This momentum-phase coupling ensures that the resulting parallel transport $U_3(e)$ depends not only on the force correlation between nodes but also on their relative kinematic state.

The regularization $\epsilon_c > 0$ ensures that $\mathbf{c}_x$ is well-defined even when the viscous force vanishes, e.g. for an isolated walker or when $\nu_{\text{visc}} = 0$. If $\mathbf{c}_x = 0$, then $\Re(\mathbf{c}_x^\dagger \lambda_a \mathbf{c}_y) = 0$ for all outgoing edges from x, hence $U_3(x \to y) = I$ for all such edges. The same holds when $\mathbf{c}_y = 0$. Nodes with trivial force therefore couple trivially through this edge transport.

The normalization satisfies $\|\mathbf{c}_x\|^2 < 1$ for all x (Lemma 5.4). Combined with $|\lambda_a|_{\text{op}} \le 2$ for all Gell-Mann generators, this gives the operator bound

$$\|\Theta_e\|_{\text{op}} \le \frac{g_3}{2} \sum_{a=1}^{8} |\Re(\mathbf{c}_x^\dagger \lambda_a \mathbf{c}_y)|\, \|\lambda_a\|_{\text{op}} \le \frac{g_3}{2} \cdot 8 \cdot \|\mathbf{c}_x\|\, \|\mathbf{c}_y\| \cdot 2 < 8g_3,$$

so the gauge coupling strength is controlled uniformly over all edges and runs.

Lemma 5.4 (Normalization of color vectors)**.** *For every node $x \in \mathcal{N}$,*

$$\mathbf{c}_x^\dagger \mathbf{c}_x = \frac{\tilde{\mathbf{c}}_x^\dagger \tilde{\mathbf{c}}_x}{\tilde{\mathbf{c}}_x^\dagger \tilde{\mathbf{c}}_x + \epsilon_c^2} \in [0, 1),$$

so $\mathbf{c}_x$ is well-defined and $\|\mathbf{c}_x\| < 1$.

Proof. Since $\epsilon_c > 0$, the denominator $\tilde{\mathbf{c}}_x^\dagger \tilde{\mathbf{c}}_x + \epsilon_c^2$ is strictly positive. The ratio $r/(r + \epsilon_c^2)$ for $r = \tilde{\mathbf{c}}_x^\dagger \tilde{\mathbf{c}}_x \geq 0$ lies in $[0, 1)$, with the upper bound strict because $\epsilon_c^2 > 0$. □

Lemma 5.5 (Color reparametrization)**.** *Let $\Omega : \mathcal{N} \to SU(3)$ be an arbitrary map. Define $\mathbf{c}_x^\Omega := \Omega(x)\mathbf{c}_x$ for each $x \in \mathcal{N}$. Then $(\mathbf{c}_x^\Omega)^\dagger \mathbf{c}_x^\Omega = \mathbf{c}_x^\dagger \mathbf{c}_x$ and $\|\mathbf{c}_x^\Omega\| = \|\mathbf{c}_x\|$.*

Proof. Since $\Omega(x) \in U(3)$, $\Omega(x)^\dagger \Omega(x) = I$. Hence

$$(\mathbf{c}_x^\Omega)^\dagger \mathbf{c}_x^\Omega = \mathbf{c}_x^\dagger \Omega(x)^\dagger \Omega(x) \mathbf{c}_x = \mathbf{c}_x^\dagger \mathbf{c}_x.$$

□

Definition 5.6 (Gell-Mann generators)**.** *The eight Gell-Mann matrices $\lambda_a \in \mathbb{C}^{3\times 3}$, $a = 1, \ldots, 8$, are*

$$\lambda_1 = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \ \lambda_2 = \begin{pmatrix} 0 & -i & 0 \\ i & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \ \lambda_3 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 0 \end{pmatrix},$$

$$\lambda_4 = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \end{pmatrix}, \ \lambda_5 = \begin{pmatrix} 0 & 0 & -i \\ 0 & 0 & 0 \\ i & 0 & 0 \end{pmatrix}, \ \lambda_6 = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix},$$

$$\lambda_7 = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & -i \\ 0 & i & 0 \end{pmatrix}, \ \lambda_8 = \frac{1}{\sqrt{3}} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -2 \end{pmatrix}.$$

Each λ_a is Hermitian and $\operatorname{tr} \lambda_a = 0$. The generators $\{i\lambda_a/2\}_{a=1}^8$ form a basis of the Lie algebra $\mathfrak{su}(3)$ of traceless skew-Hermitian matrices [2].

Lemma 5.7 (Gell-Mann orthogonality)**.** *The Gell-Mann matrices satisfy*

$$\operatorname{tr}(\lambda_a \lambda_b) = 2\delta_{ab}, \qquad a, b \in \{1, \ldots, 8\}.$$

Proof. Direct matrix multiplication: for $a = b$ each diagonal entry of λ_a^2 contributes, giving trace 2; for $a \neq b$ the off-diagonal structure ensures the trace vanishes. This is a finite componentwise check whose result is well-known [2]. □

Lemma 5.8 (Hermitian traceless span)**.** *If $\alpha_1, \ldots, \alpha_8 \in \mathbb{R}$, then $H := \sum_{a=1}^8 \alpha_a \lambda_a$ is Hermitian and traceless.*

Proof. Each λ_a is Hermitian and traceless by Definition 5.6. Real linear combinations preserve both properties. □

Definition 5.9 (Generator and transport on edges)**.** *For each oriented edge $e = (x \to y) \in \mathcal{E}$ define the algebra element*

$$\mathcal{A}_e := \frac{1}{2}\sum_{a=1}^{8} \Re\big(\mathbf{c}_x^\dagger \lambda_a \mathbf{c}_y\big)\lambda_a \in \{H \in \mathbb{C}^{3\times 3} : H^\dagger = H,\ \operatorname{tr} H = 0\}, \qquad \Theta_e := g_3\, \mathcal{A}_e.$$

and the $SU(3)$ edge transport

$$U_3(e) := \exp(i\Theta_e) \in SU(3), \qquad U_3(e^{-1}) := U_3(e)^{-1}.$$

Remark 5.10 (Justification of the transport construction)*.* The formula

$$\mathcal{A}_e = \frac{1}{2}\sum_a \Re\big(\mathbf{c}_x^\dagger \lambda_a \mathbf{c}_y\big)\lambda_a$$

is the orthogonal projection of the color bilinear $\mathbf{c}_x\mathbf{c}_y^\dagger$ onto the Hermitian traceless subspace spanned by $\{\lambda_a\}_{a=1}^8$. To see this, recall that $\mathfrak{su}(3)$ carries the Killing-normalized inner product $\langle A, B\rangle = \frac{1}{2}\operatorname{tr}(AB)$, under which $\{\lambda_a\}$ is an orthogonal basis with $\langle \lambda_a, \lambda_b\rangle = \delta_{ab}$ by Lemma 5.7. The Hermitian part of any matrix M is $(M + M^\dagger)/2$. The a-th component of the projection of $\mathbf{c}_x\mathbf{c}_y^\dagger$ is

$$\big\langle \lambda_a, \tfrac{1}{2}(\mathbf{c}_x\mathbf{c}_y^\dagger + \mathbf{c}_y\mathbf{c}_x^\dagger)\big\rangle = \frac{1}{2}\operatorname{tr}\big(\lambda_a(\mathbf{c}_x\mathbf{c}_y^\dagger + \mathbf{c}_y\mathbf{c}_x^\dagger)\big) = \Re\big(\mathbf{c}_y^\dagger\lambda_a\mathbf{c}_x\big) = \Re\big(\mathbf{c}_x^\dagger\lambda_a\mathbf{c}_y\big)^*,$$

where the last equality uses the Hermitian property $\lambda_a^\dagger = \lambda_a$. Since the resulting coefficients are real, taking $\Re(\cdot)$ is consistent, and $\mathcal{A}_e$ is exactly this projection. The coupling is $\Theta_e = g_3\mathcal{A}_e$.

The factor $g_3/2$ is chosen to match the continuum covariant derivative $D_\mu = \partial_\mu - ig_3A_\mu^a\lambda_a/2$: in the weak-coupling limit $g_3 \to 0$ the transport linearizes to

$$U_3(e) = I + i\Theta_e + O(g_3^2) = I + ig_3\,\mathcal{A}_e + O(g_3^2) = I + \frac{ig_3}{2}\sum_a \Re(\mathbf{c}_x^\dagger\lambda_a\mathbf{c}_y)\lambda_a + O(g_3^2),$$

which is the first-order approximation to a continuum parallel transporter $P\exp(ig_3\int A_\mu\, dx^\mu)$ [4].

Proposition 5.11 (Edge transport is $SU(3)$)**.** *For every $e \in \mathcal{E}$, $U_3(e) \in SU(3)$ and $U_3(e^{-1}) = U_3(e)^{-1}$.*

Proof. Each coefficient $\Re(\mathbf{c}_x^\dagger\lambda_a\mathbf{c}_y)$ is real. By Lemma 5.8, Θ_e is Hermitian and traceless, so $i\Theta_e$ is skew-Hermitian and traceless. Hence $U_3(e) = \exp(i\Theta_e)$ is unitary. Since $\operatorname{tr}\Theta_e = 0$, the identity $\det\exp(M) = \exp(\operatorname{tr} M)$ gives $\det U_3(e) = \exp(i\operatorname{tr}\Theta_e) = 1$. Therefore $U_3(e) \in SU(3)$. The inverse relation $U_3(e^{-1}) = U_3(e)^{-1}$ is imposed by convention. □

Remark 5.12. There are no oriented self-loops in $\mathcal{E}$: all edges are either time-shifted, same-time or influence-ahead, so source and target are distinct by definition. If one formally assigns the transport formula to a self-loop, it still lands in $SU(3)$ because $\mathcal{A}_e$ is Hermitian and traceless.

6 Holonomy and gauge invariance

Definition 6.1 (Path holonomy)**.** *For an oriented path $\gamma = (v_0 \to v_1, \ldots, v_{m-1} \to v_m)$ with $v_k \in \mathcal{N}$, define the ordered product (left-to-right composition)*

$$U_3(\gamma) := U_3(v_0 \to v_1)\, U_3(v_1 \to v_2) \cdots U_3(v_{m-1} \to v_m) = \prod_{k=1}^{m} U_3(v_{k-1} \to v_k).$$

This ordering convention matches the standard lattice QCD convention in which $U_\mu(x)$ transports from site x to site $x+\hat{\mu}$ [4]. If $v_m = v_0$, γ is a loop and we define the Wilson loop trace $W_3(\gamma) := \operatorname{tr} U_3(\gamma)$.

Definition 6.2 (Triangle and plaquette holonomy)**.** *For a triangle or plaquette with oriented boundary edges, define*

$$U_3(\triangle) := \prod_{e\in\partial\triangle} U_3(e), \qquad U_3(P) := \prod_{e\in\partial P} U_3(e),$$

with factors ordered according to the oriented boundary sequence.

Proposition 6.3 (Plaquette factorization)**.** *For each plaquette $P_{ij,t}$,*

$$U_3(P_{ij,t}) = U_3(\triangle_{ij,t})\, U_3(\triangle_{ji,t}).$$

Proof. The boundary of $P_{ij,t}$ is by definition $\partial P_{ij,t} = \partial\triangle_{ij,t} + \partial\triangle_{ji,t}$, with the IG edges appearing with opposite orientation. Specifically, $\partial\triangle_{ij,t}$ contains the edge $e_{\mathrm{IG}}(n_{i,t} \to n_{j,t})$ with coefficient -1, meaning the holonomy for $\triangle_{ij,t}$ includes the factor $U_3(e_{\mathrm{IG}}(n_{i,t} \to n_{j,t}))^{-1}$. Similarly, $\partial\triangle_{ji,t}$ contains $e_{\mathrm{IG}}(n_{j,t} \to n_{i,t})$ with coefficient -1, giving the factor $U_3(e_{\mathrm{IG}}(n_{j,t} \to n_{i,t}))^{-1} = U_3(e_{\mathrm{IG}}(n_{i,t} \to n_{j,t}))$. The product of these two contributions is $U_3(e)^{-1} \cdot U_3(e) = I$, so the IG edge cancels and $U_3(P_{ij,t}) = U_3(\triangle_{ij,t})\, U_3(\triangle_{ji,t})$. □

Definition 6.4 (Gauge map)**.** *A gauge map is any map $\Omega : \mathcal{N} \to SU(3)$. For an edge $e = (x \to y) \in \mathcal{E}$ define the gauge-transformed transport*

$$U_3^{\Omega}(e) := \Omega(y)\, U_3(e)\, \Omega(x)^{-1}.$$

Transformed color vectors are $\mathbf{c}_x^{\Omega} := \Omega(x)\mathbf{c}_x$.

Proposition 6.5 (Gauge invariance of Wilson traces)**.** *For any closed loop $\gamma = (v_0, \ldots, v_m = v_0)$ and any gauge map Ω,*

$$\operatorname{tr}\left(U_3^{\Omega}(\gamma)\right) = \operatorname{tr}\left(U_3(\gamma)\right).$$

Proof. Compute the gauge-transformed holonomy:

$$U_3^{\Omega}(\gamma) = \prod_{k=1}^{m} \Omega(v_k)\, U_3(v_{k-1} \to v_k)\, \Omega(v_{k-1})^{-1}.$$

Consecutive factors telescope: $\Omega(v_k)$ from step k cancels $\Omega(v_k)^{-1}$ from step $k+1$, since v_k appears as the target of step k and the source of step $k+1$. After all m steps, the remaining factors are $\Omega(v_m) = \Omega(v_0)$ on the left and $\Omega(v_0)^{-1}$ on the right, giving

$$U_3^{\Omega}(\gamma) = \Omega(v_0)\, U_3(\gamma)\, \Omega(v_0)^{-1}.$$

Trace cyclicity then gives $\operatorname{tr}(U_3^{\Omega}(\gamma)) = \operatorname{tr}(\Omega(v_0)^{-1}\Omega(v_0)\, U_3(\gamma)) = \operatorname{tr}(U_3(\gamma))$. □

Definition 6.6 (Wilson action)**.** *Let*

$$\mathcal{P}_{\square} := \{P_{ij,t} \mid (i,j),(j,i) \in \mathcal{P}_t,\ t = 0, \ldots, T-1\}$$

be the set of all plaquettes, and set $\beta := 6/g_3^2$. The Wilson plaquette action is

$$S_{\mathrm{W}} := \beta \sum_{P\in\mathcal{P}_{\square}} \left(1 - \frac{1}{3}\Re \operatorname{tr} U_3(P)\right).$$

Remark 6.7. The normalization $1 - \frac{1}{3}\Re\,\mathrm{tr}\,U_3(P)$ vanishes when $U_3(P) = I$ and is non-negative for all $U_3(P) \in SU(3)$, since $|\,\mathrm{tr}\,U|/3 \le 1$ by the triangle inequality applied to the three eigenvalues of U. In the weak-coupling limit $g_3 \to 0$, $U_3(P) \approx I + ig_3^2 F_P + O(g_3^4)$ for a field-strength matrix F_P, and the action approaches $\frac{g_3^2}{6} \cdot 6/g_3^2\,\mathrm{tr}(F_P^2)/2 = \frac{1}{2}\,\mathrm{tr}(F_P^2)$, recovering the continuum Yang-Mills action [5, 1].

Proposition 6.8 (Gauge invariance of Wilson action)**.** *For any gauge map Ω, $S_{\mathrm{W}}^{\Omega} = S_{\mathrm{W}}$.*

Proof. Each plaquette trace $\Re\,\mathrm{tr}\,U_3(P)$ is invariant under gauge transformation by Proposition 6.5. The sum and the constant term are unchanged, so $S_{\mathrm{W}}^{\Omega} = S_{\mathrm{W}}$. □

7 Matter sector on the same complex

Definition 7.1 (Matter fields and actions)**.** *For each node $x \in \mathcal{N}$ choose a fermionic spinor $\psi_x \in \mathbb{C}^3$, its conjugate $\bar{\psi}_x \in (\mathbb{C}^3)^\dagger$, and a real scalar $\phi_x \in \mathbb{R}$. Fix $m_F > 0$, $m_\phi > 0$, and a local potential $V_\phi : \mathbb{R} \to \mathbb{R}$. For $e = (x \to y) \in \mathcal{E}$ set $\ell_{xy} := \|\mathbf{x}(y) - \mathbf{x}(x)\| + \eta$ with $\eta > 0$. The covariant fermionic action is*

$$S_{\mathrm{F}} := \sum_{e \in E_{\mathrm{CST}} \cup E_{\mathrm{IA}}} \bar{\psi}_{t(e)} \left(\psi_{t(e)} - U_3(e)\psi_{s(e)}\right) + m_F \sum_{x \in \mathcal{N}} \bar{\psi}_x \psi_x,$$

and the covariant scalar action is

$$S_\phi := \frac{1}{2} \sum_{e \in E_{\mathrm{CST}} \cup E_{\mathrm{IG}}} \frac{(\phi_{t(e)} - \phi_{s(e)})^2}{\ell_{xy}} + \sum_{x \in \mathcal{N}} \Big(\frac{m_\phi^2}{2}\phi_x^2 + V_\phi(\phi_x)\Big).$$

The gauge group acts on matter fields by

$$\psi_x \mapsto \Omega(x)\psi_x, \qquad \bar{\psi}_x \mapsto \bar{\psi}_x \Omega(x)^{-1}, \qquad \phi_x \mapsto \phi_x.$$

Remark 7.2 (Distance weighting in the scalar action)*.* The factor $1/\ell_{xy}$ in the scalar kinetic term is a discrete analogue of the continuum normalization $(\nabla_\mu \phi)^2$. In the continuum, the square of a derivative has dimensions $[\phi]^2/[\text{length}]^2$, so a finite difference $(\phi_y - \phi_x)^2$ must be divided by $\|\mathbf{x}(y) - \mathbf{x}(x)\|$ (one power of length rather than two) for a first-order finite-difference scheme. Specifically, if $\phi_{t(e)} - \phi_{s(e)} \approx (\nabla\phi) \cdot (\mathbf{x}(y) - \mathbf{x}(x))$ with $|\mathbf{x}(y) - \mathbf{x}(x)| \approx \ell_{xy}$, then $(\phi_{t(e)} - \phi_{s(e)})^2/\ell_{xy} \approx \ell_{xy}\,|\nabla\phi|^2$, and summing over a regular lattice with spacing $a = \ell_{xy}$ recovers $a^d\,|\nabla\phi|^2$ as in the standard scalar lattice action [3].

The parameter $\eta > 0$ prevents division by zero for coincident node positions (self-loops, or walkers occupying the same point at the same time). In practice $\eta \ll \ell_{\mathrm{visc}}$ so that $\ell_{xy} \approx \|\mathbf{x}(y) - \mathbf{x}(x)\|$ for all non-coincident edges.

Proposition 7.3 (Matter gauge invariance)**.** *Under the gauge action of Definition 7.1,*

$$S_{\mathrm{F}}^{\Omega} = S_{\mathrm{F}}, \qquad S_\phi^{\Omega} = S_\phi.$$

Proof. For each fermionic edge term,

$$\begin{aligned}\bar{\psi}_{t(e)}\big(\psi_{t(e)} - U_3(e)\psi_{s(e)}\big) &\mapsto \bar{\psi}_{t(e)}\Omega(t(e))^{-1}\big(\Omega(t(e))\psi_{t(e)} - \Omega(t(e))U_3(e)\Omega(s(e))^{-1}\Omega(s(e))\psi_{s(e)}\big)\\ &= \bar{\psi}_{t(e)}\big(\psi_{t(e)} - U_3(e)\psi_{s(e)}\big)\,,\end{aligned}$$

where we used $\Omega(t(e))^{-1}\Omega(t(e)) = I$ and $\Omega(s(e))^{-1}\Omega(s(e)) = I$. The mass term $m_F\bar{\psi}_x\psi_x$ is invariant because $\Omega(x)^{-1}\Omega(x) = I$ pointwise. The scalar action S_ϕ depends only on scalar fields, which are gauge-invariant, so $S_\phi^{\Omega} = S_\phi$. □

Remark 7.4 (Edge-type selection in the matter actions). The fermionic action uses $E_{\mathrm{CST}} \cup E_{\mathrm{IA}}$ rather than the full edge set $\mathcal{E}$. CST edges encode the temporal forward propagation of each walker: a fermion at node $n_{i,t}$ propagates causally to $n_{i,t+1}$. IA edges encode the causal influence of a companion j at time t on walker i at the later time $t+1$: this is precisely the interaction-ahead coupling. Together, $E_{\mathrm{CST}} \cup E_{\mathrm{IA}}$ spans all edges that connect a node to a causally later node, which is the natural support for a first-order discrete Dirac operator.

The scalar action uses $E_{\mathrm{CST}} \cup E_{\mathrm{IG}}$ instead. Scalars have no preferred propagation direction and can couple instantaneously within a timestep; the IG edges at the same time t provide exactly this same-time stiffness. Using IA edges in the scalar action would couple $\phi_{n_{i,t+1}}$ to $\phi_{n_{j,t}}$ across a time boundary, which is less natural for a scalar and would break the symmetry between i and j at time t.

8 Finite lattice-QFT data package

Theorem 8.1 (Finite $SU(3)$ data package)**.** *Given a run $\mathscr{R}$ as in Definition 3.1 and the constructions in Sections 3–7, the tuple*

$$\mathcal{Q} = \Big(T, \{\mathcal{A}(t)\}, \{\mathcal{P}_t\}, \mathcal{N}, \mathcal{E}, \mathcal{T}, \{x_i(t), v_i(t), \Phi_i(t)\}_{i,t}, K_\rho, \nu_{\mathrm{visc}}, \hbar_{\mathrm{eff}}, \mu_p, \ell_0, \epsilon_c, g_3, \beta, \eta, \{U_3(e)\}_{e\in\mathcal{E}}, S_{\mathrm{W}}, S_{\mathrm{F}}, S_\phi\Big)$$

is finite and explicit. It carries:

1. *a directed 2-complex $(\mathcal{N}, \mathcal{E}, \mathcal{T})$ with boundary operators (∂_1, ∂_2) satisfying $\partial_1\partial_2 = 0$;*
2. *$SU(3)$ parallel transport maps $\{U_3(e)\}_{e\in\mathcal{E}}$;*
3. *gauge-invariant Wilson traces and Wilson action S_{W};*
4. *gauge-covariant fermionic and scalar matter actions S_{F}, S_ϕ.*

All objects required for a subsequent axiom-verification paper are present.

Proof. Finiteness of $(\mathcal{N}, \mathcal{E}, \mathcal{T})$ is Proposition 3.6. Finiteness of $\{U_3(e)\}$ follows because $\mathcal{E}$ is finite and each $U_3(e)$ is defined by a finite matrix exponential. The actions S_{W}, S_{F}, S_ϕ are finite sums over finite index sets by the same finiteness. Gauge invariance of S_{W} is Proposition 6.8; gauge invariance of S_{F} and S_ϕ is Proposition 7.3. The boundary identity $\partial_1\partial_2 = 0$ is Proposition 3.5. □

9 Discussion

9.1 Relationship to standard lattice QCD

Wilson's original lattice gauge theory [5] assigns an independent $SU(3)$-valued link variable $U_\mu(x)$ to each directed bond of a regular hypercubic spacetime lattice. The link variables are the fundamental integration variables of the path integral, and the gauge group is imposed as an exact symmetry from the outset. The present construction differs in two respects.

First, the underlying complex $(\mathcal{N}, \mathcal{E}, \mathcal{T})$ is not a regular lattice but an irregular, data-dependent graph whose topology changes from run to run. The node set is a finite subset of a spacetime determined by the alive walkers, and the edge set encodes three physically distinct types of coupling: temporal propagation (CST), same-time interaction (IG), and causal interaction-ahead (IA). In lattice QCD all bonds are topologically equivalent; here they are not.

Second, the link variables $U_3(e)$ are not independent: they are *derived* from the trajectory data via the color construction of Definitions 5.1–5.9. The gauge group, the coupling strength g_3, and the parallel transport all emerge from the viscous dynamics rather than being postulated. This derivation-vs-postulation distinction is the central new feature of the construction.

Despite these differences, the Wilson action $S_{\mathrm{W}} = \beta \sum_P (1 - \frac{1}{3}\Re \operatorname{tr} U_3(P))$ has exactly the standard lattice QCD form [1, 4], and all the gauge-theoretic machinery (holonomies, Wilson loops, covariant derivatives) is identical to the standard treatment. This means that any lattice QCD observable defined in terms of the link variables can be computed on $\mathcal{Q}$ without modification.

9.2 Continuum limit

In standard lattice QCD one takes the continuum limit by sending the lattice spacing $a \to 0$ while tuning g to keep physical masses fixed (asymptotic freedom). An analogous program for the interaction complex would require:

1. a sequence of runs with increasing T and N such that the empirical spatial density N/V diverges and the time step $\Delta t \to 0$;
2. a rescaling of the coupling $g_3 \sim a(\log a)^{-1}$ and the de Broglie parameters with the effective lattice spacing $a \sim \ell_{\mathrm{visc}}/N^{1/3}$;
3. convergence of the empirical measure $\frac{1}{N}\sum_i \delta_{x_i(t)}$ to a smooth spatial density as $N \to \infty$, as supplied by the kinetic convergence program in the companion paper.

This program requires results from the kinetic convergence theory developed in the companion paper and is outside the present scope.

9.3 Quantization and partition function

The data package $\mathcal{Q}$ provides a well-defined action $S = S_{\mathrm{W}} + S_{\mathrm{F}} + S_\phi$ on the finite-dimensional field variables $\{\psi_x, \bar{\psi}_x, \phi_x\}_{x\in\mathcal{N}}$. The path integral over these fields,

$$Z = \int \Big[\prod_{x\in\mathcal{N}} d\psi_x \, d\bar{\psi}_x \, d\phi_x\Big] \exp\!\big(-S_{\mathrm{W}} - S_{\mathrm{F}} - S_\phi\big),$$

is a finite-dimensional ordinary integral over the gauge sector $SU(3)^{|\mathcal{N}|}$, finite-dimensional fermionic Grassmann variables, and the scalar sector $\mathbb{R}^{|\mathcal{N}|}$. Gauge-fixing via Faddeev–Popov is straightforward in the finite setting [3]. The resulting Z is well-defined without any ultraviolet regularization; all divergences that would appear in a continuum QFT are absent because the complex is finite. Evaluating Z numerically for realistic run data and comparing with QCD predictions is a natural next step.

10 Conclusions

We have constructed a complete, finite, and $SU(3)$ gauge-invariant lattice QFT data package $\mathcal{Q}$ from the discrete trajectory data of a viscous multi-particle system. The construction proceeds in five conceptual steps:

1. Build the directed 2-complex $(\mathcal{N}, \mathcal{E}, \mathcal{T})$ from the spacetime nodes, causal edges, interaction edges, and interaction-ahead edges of the run.

2. Compute the row-normalized viscous force $\mathbf{F}^{\mathrm{visc}}(x)$ at each node from the interaction graph weights.

3. Form complex color vectors $\mathbf{c}_x \in \mathbb{C}^3$ by modulating each force component with a de Broglie kinematic phase and normalizing.

4. Define the $SU(3)$ edge transport $U_3(e) = \exp(i\Theta_e)$ by projecting the bilinear color overlap $\mathbf{c}_x^\dagger \lambda_a \mathbf{c}_y$ onto the Lie algebra $\mathfrak{su}(3)$ via the Gell-Mann basis.

5. Construct the Wilson plaquette action S_{W} and the covariant matter actions S_{F}, S_ϕ on the same complex.

All objects are explicit and finite (Theorem 8.1). The gauge group $SU(3)$ is not imposed but emerges from the three-dimensional viscous dynamics together with the Gell-Mann projection.

Three major extensions are deferred to companion papers: (i) the formal continuum limit as run data becomes dense; (ii) path-integral quantization and numerical evaluation of Z; (iii) axiom verification — checking that $\mathcal{Q}$ satisfies the Osterwalder-Schrader axioms adapted to the irregular complex.

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